Topic 6: Organic Chemistry I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include cracking of artificial crude oil, extracting limonene from orange peel, dehydrating an alcohol to an alkene, preparing a simple halogenoalkane such as 1-bromobutane, simple test tube reactions for different functional groups.

Mathematical skills that could be developed in this topic include calculating the yield of a reaction or an atom economy.

Within this topic, students can consider how the polymer industry provides useful solutions for many modern applications, but poses questions about sustainability of resources and the feasibility of recycling. They will also encounter practical organic chemistry, showing them how chemists work safely with potentially hazardous chemicals by managing risks.

Students should:	
Topic 6A: Introduction to organic chemistry	
1.	know that a hydrocarbon is a compound of hydrogen and carbon only
2.	be able to represent organic molecules using empirical formulae, molecular formulae, general formulae, structural formulae, displayed formulae and skeletal formulae
3.	know what is meant by the terms 'homologous series' and 'functional group'
4.	be able to name compounds relevant to this specification using the rules of International Union of Pure and Applied Chemistry (IUPAC) nomenclature *Students will be expected to know prefixes for compounds up to* C_{10}
5.	be able to classify reactions as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation
6.	understand the term 'structural isomerism' and determine the possible structural, displayed and skeletal formulae of an organic molecule, given its molecular formula
7.	understand the term 'stereoisomerism', as illustrated by *E/Z* isomerism (including *cis-trans* isomerism where two of the substituent groups are the same)